InmoScore - Final Report

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Delivery 2 · Final Submission
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1. Problem Statement

1.1 Background

Finding a home in Lima Metropolitana is a complex, time-consuming, and often frustrating process. Potential buyers face several challenges:

- **Information Overload**: Multiple platforms (Urbania, MercadoLibre, real estate agencies) with inconsistent data formats
- **Lack of Personalization**: Listings are sorted by price or date, not by individual buyer preferences
- **Unclear Value**: Buyers struggle to determine if a listed price is fair for the market
- **Opaque Decision Factors**: Limited visibility into how location, safety, and convenience factors into property value
- **Financial Uncertainty**: Buyers don't know if they can afford specific properties given their income and savings

1.2 Problem

Home buyers in Lima lack a transparent, personalized tool that:

- Explains why properties are recommended
- Accounts for individual financial constraints
- Provides objective price references
- Considers location quality indicators (safety, convenience)
- Allows easy comparison of multiple options

1.3 Input/Output

Input:

- Property listings (price, location, attributes)
- Buyer profile (income, savings, budget, preferences)
- District-level indicators (crime index, urban convenience, poverty rate)

Output:

- Personalized property ranking with scores
- Price references vs. market estimates
- Opportunity indicators (price below estimated value)
- Comparison tools for multiple properties

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2. Target Users and Value Proposition

2.1 Target Users

Primary User: First-time home buyers in Lima Metropolitana

- Age: 25-45
- Income: S/ 5,000 - S/ 30,000 monthly
- Goal: Purchase their first property for residence
- Pain points: Uncertainty about fair pricing, overwhelmed by options, need for location safety

Secondary Users:

- Real estate agents: Quick reference for client preferences
- Repeat buyers: Comparison tool for investment properties

2.2 Value Proposition

InmoScore provides:

- **Transparency**: Clear explanation of why properties are recommended
- **Personalization**: Rankings based on individual financial situation and priorities
- **Objective Reference**: Machine learning price estimates as market benchmarks
- **Decision Support**: Side-by-side comparison of multiple properties
- **Financial Clarity**: Affordability calculations based on income and savings

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3. Data Sources

3.1 Primary Dataset

Source: Urbania Lima Real Estate Dataset (publicly available on GitHub)

- **Repository:** `MathiuCz/lima-real-estate-price-predictor`
- **License:** MIT (repository license; content subject to Urbania terms)
- **Original Collection:** Scraped from Urbania real estate platform
- **Usage:** Academic demonstration only; not licensed for commercial republication

3.2 Dataset Characteristics

Attribute	Value
Historical base listings	3,896
Listings in final application catalog	6,340 (historical base plus dated Urbania captures; 37 listings observed on 2026-11-08)
Districts	20 districts in Lima
Time period	Historical listings plus captures from 2026-10-20 through 2026-11-08; only 37 catalogue listings were observed on the latest date
Fields	40 columns including price, area, rooms, bathrooms, location

3.3 Additional Data Sources

Source	Variable	Purpose
SIDPOL/MININTER	Crime index by district	Safety indicator
OpenStreetMap	Geocoding (133 listings)	Location precision
BCRP	Economic indicators	Context (not directly used)
INEI	IDH, poverty rates	District context
Fondo MIVIVIENDA	Housing projects	Urban development context

3.4 Data Limitations

- **Historical Data**: Listings are not verified for current availability
- **No Temporal Validation**: Cannot assess price trends over time
- **Incomplete Coverage**: Only 133 of 6,340 catalogue listings (2.1%) have precise geocoding
- **Selection Bias**: 91 listings excluded by IQR filtering in Week 7
- **Maintenance Unknown**: 4,453 of 6,340 catalog listings (70.24%) lack maintenance fee data; 2,009 belong to the historical base

4. Data Preparation

4.1 Preprocessing Pipeline

The data preparation pipeline (`analytics/prepare_data.py`) performs:

1. **Data Loading**: Load `listings_clean.csv` and the dated capture
2. **Context Loading**: Read district indices and validated geocoding
3. **Capture Merge**: Update 567 historical IDs and add 2,444 usable new IDs from the accumulated dated captures
4. **Model Preprocessing**: Preserve numeric missing values for HistGradientBoosting and one-hot encode district using the fitted pipeline
5. **Model Application**: Apply trained Gradient Boosting model to all listings
6. **Catalog Generation**: Create `catalog.json` with predictions and scores

4.2 Feature Selection

Features used for price prediction:

- Property attributes: area_m2, rooms, bathrooms, parking, maintenance_fee
- District context: crime_index_zone, urban_convenience_index, zone_composite_index
- Socioeconomic: idh_2019, pct_pobreza_total
- Development: proyectos_nuevos_distrito, unidades_nuevas_distrito
- Categorical: district (one-hot encoded)

4.3 Data Quality Notes

- **Missing Values**: The selected HistGradientBoosting model handles numeric missing values. IQR filters training outliers; it is not an imputation method.
- **Outliers**: 91 listings excluded by IQR filtering in historical evaluation
- **Duplicates**: 24 potential duplicates identified with link lookup
- **Geocoding**: 133 of 6,340 catalogue listings (2.1%) have validated street/building-level coordinates; no point is shown for the remaining listings

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5. Requirements

5.1 Functional Requirements

ID	Requirement	Priority
FR-1	User registration with email and password	High
FR-2	Profile creation with income, savings, budget, preferences	High
FR-3	Personalized property ranking based on profile	High
FR-4	Price reference vs. model estimate	High

ID	Requirement	Priority
FR-5	Save favorites to account	Medium
FR-6	Compare up to 3 properties	Medium
FR-7	Location information with OpenStreetMap links	Medium
FR-8	Mobile-responsive interface	High
FR-9	Account deletion with confirmation	Low

5.2 Non-Functional Requirements

ID	Requirement	Priority
NFR-1	Response time < 2s for ranking	High
NFR-2	Secure password storage (scrypt)	High
NFR-3	Session management with HttpOnly cookies	High
NFR-4	SQLite local persistence for prototype	Medium
NFR-5	Data validation on all inputs	High
NFR-6	Graceful error handling	Medium

5.3 Constraints

- **Technical**: Node.js 22.13+ required (uses experimental `node:sqlite`)
- **Legal**: Cannot republish Urbania data commercially
- **Data**: Limited to historical listings; no real-time verification
- **Scope**: Prototype only; not production-ready for commercial use

5.4 Acceptance Criteria

- User can register and create profile
- Ranking returns relevant properties within budget
- Price estimates are reasonable (MAE < S/ 150,000)
- Favorites persist across sessions
- Mobile view is functional on 320px+ screens
- Account deletion removes all user data

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6. Modeling and Analytical Approach

6.1 Model Selection

Selected Model: Gradient Boosting (Model D)

- **Framework:** scikit-learn HistGradientBoostingRegressor

- **Objective:** Predict log(price) to stabilize variance
- **Features:** 13 features including property attributes and district context

6.2 Baseline Comparison

Selection used only pooled out-of-fold predictions on the 3,117 development listings:

Model	Development MAE (S/)	Development MdAPE	MAE reduction vs baseline
A: Median by district	154,449	15.08%	0%
B: Log-linear with district	153,155	13.66%	0.84%
C: Ridge v2	142,694	13.66%	7.61%
D: Gradient Boosting	135,553	12.04%	12.23%

After selection, only the baseline and selected model were evaluated on the 779 reserved final listings:

Model	Final-test MAE (S/)	Final-test MdAPE	MAE reduction vs baseline
A: Median by district	153,665	16.11%	0%
D: Gradient Boosting	119,801	12.18%	22.04%

Gradient Boosting was selected using development performance and the small-district guardrail. The final-test result confirms the comparison for the historical snapshot; it was not used for selection.

6.3 Evaluation Strategy

- **Dataset Split:** 3,117 development and 779 final test, grouped by possible duplicates. Training-only IQR removes 19 development rows; the exported model is fitted on 3,098 rows.
- **Cross-Validation:** 5-fold CV on training set
- **Metrics:** MAE (absolute error), MdAPE (percentage error)
- **Segment Analysis:** Performance evaluated by price segments

6.4 Model Limitations

- **High-Price Segment**: For properties ≥ S/ 2M (36 listings), Gradient Boosting performs 2.6% worse than baseline
- **Historical Bias**: Model trained on historical data; may not reflect current market
- **Temporal Validation**: Not possible due to lack of publication dates
- **Geographic Scope**: Limited to 20 districts in dataset

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7. Product Design

7.1 System Architecture

```
User Browser
    ↓ (HTTP/HTTPS)
Next.js Application (app/, components/)
    ↓ (API Routes)
API Layer (app/api/)
    ↓ (SQLite)
Account API -> Repository -> SQLite (profiles, favorites, sessions)
Ranking API -> Read-only catalogue (data/catalog.json)
    ↑ (Generated by)
Analytics Pipeline (analytics/prepare_data.py)
    ↑ (Uses)
Model (models/valuation_model.joblib)
```

7.2 User Flow

1. **Landing**: Understand product value proposition
2. **Registration**: Create account with email/password
3. **Profile Setup**: Define income, savings, budget, preferences
4. **Ranking**: View personalized property recommendations
5. **Detail**: Explore individual property with price reference
6. **Save/Compare**: Save favorites, compare multiple options
7. **Account Management**: Edit profile, view favorites, delete account

7.3 Interface Design Principles

- **Clarity**: Explain recommendations with reasons
- **Simplicity**: Minimal steps to get recommendations
- **Transparency**: Show how scores are calculated
- **Responsiveness**: Works on mobile, tablet, desktop
- **Accessibility**: Semantic HTML, keyboard navigation, focus indicators

7.4 Technical Stack

- **Frontend**: Next.js 16.3.4, React 19, TypeScript
- **Styling**: Tailwind CSS 4.2.1
- **Database**: SQLite (node:sqlite experimental feature)
- **ML**: scikit-learn (Python)
- **Authentication**: Custom implementation with scrypt

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8. Prototyping Process

8.1 Evolution Timeline

Week	Milestone	Key Deliverables
Week 4	Topic & Dataset Selection	README.md, data sample, data dictionary
Week 5	Project Proposal	ProjectProposal.pdf, DataProductCanvas.pdf, Requirements.md
Week 6	EDA & Model Selection	DataAnalysis.md, ModelSelection.md
Week 7	Delivery 1	Integrated definition, architecture, wireframes
Week 10	Functional Prototype	Working prototype with main flow
Week 12	Refined Prototype	Improved functionality, evaluation, case studies
Week 15	Final Submission	Complete product, documentation, presentation

8.2 Key Decisions

- **Week 7**: Switched from HTML prototype to Next.js for better scalability
- **Week 10**: Implemented SQLite for local persistence (vs. in-memory)
- **Week 12**: Applied professor's corrections (validation, error handling, district indicators)
- **Week 14**: Refined product without adding redundant features

8.3 Technical Challenges

1. **SQLite in Node.js**: Required Node.js 22.13+ for experimental `node:sqlite`
2. **Model Deployment**: Chose offline batch processing vs. real-time inference
3. **Authentication**: Custom implementation vs. third-party services
4. **Geocoding**: Limited to 133 listings with precise coordinates
5. **Mobile Responsiveness**: Required extensive CSS media queries

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9. Evaluation

9.1 Model Performance

Final Test Set (779 listings):

- **MAE**: S/ 119,801
- **MdAPE**: 12.2%
- **Improvement vs. baseline**: 22.0%

Interpretation: The median absolute percentage error on the historical final test is 12.2%. This aggregate statistic is not a confidence interval or an error guarantee for an individual property.

9.2 Segment Analysis

Price Segment	n	MAE (S/)	Notes
All prices	779	119,801	Overall performance
≥ S/ 2M	36	663,562	2.6% worse than baseline
< S/ 2M	743	93,454	Weighted MAE from the four saved price segments

Conclusion: Model performs poorly on luxury segment due to limited training data.

9.3 Functional Testing

- **Unit Tests**: 18 tests (9 ranking, 4 auth, 5 database)
- **API Tests**: 33 API validation tests
- **Data Validation**: Automated checks via `analytics/validate_data.py`
- **Responsive Testing**: Verified on 320px, 390px, 768px, 1440px, 2560px, 3840px

9.4 Usability Observations

- **Strengths**: Clear ranking explanations, intuitive profile setup
- **Weaknesses**: District selection dropdown could be improved with search
- **Opportunities**: Add saved filters, notification of new matching listings

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10. Results

10.1 Product Capabilities

InmoScore successfully delivers:

- **Personalized Rankings**: 6,340 catalog listings can be ranked by individual preferences
- **Price References**: Model estimates provide objective market benchmarks
- **Affordability Calculations**: Users understand their purchasing power
- **Comparison Tools**: Side-by-side comparison of up to 3 properties
- **Persistence**: User accounts save favorites and profiles across sessions

10.2 User Experience

The implemented flow supports profile setup, filtered ranking, price references, favorites, comparison and account deletion. Task completion times and comprehension have not yet been measured with buyers. Functional checks do not establish usability outcomes.

10.3 Technical Achievements

- **Model**: 22% error reduction vs. baseline
- **Performance**: The catalogue is precalculated; this review does not establish a production latency or load benchmark
- **Reliability**: Error handling for partial failures (favorites)

- **Security**: Passwords hashed with scrypt (N=32768, r=8, p=3)
- **Accessibility**: Semantic HTML, keyboard navigation, focus indicators

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11. Reproducible Demonstration Case

This is a synthetic demonstration executed on the included catalogue, not a buyer study or evidence of a property visit.

Profile: María from the interface: income S/ 15,000/month, savings S/ 150,000, declared budget S/ 700,000, at least three bedrooms and 60 m², no district restriction. Initial relative weights: price 0.20, safety 0.45, convenience 0.35.

Configuration	Calculated cap (S/)	First listing ID	District	Asking price (S/)	Score
Initial profile	662,958	150180312	San Martin De Porres	148,000	94
Only price weight changes to 0.80	662,958	148659917	San Martin De Porres	149,999	96

Both configurations return 60 options after the maximum of three per district is applied in affinity ordering. Changing one weight changes the order, while the financial cap stays constant. These low asking prices must be checked with the source; model gaps are not evidence of a bargain or current availability. Affinity is a weighted score, not a satisfaction probability.

The profile named María in `data/demo_profiles.json` has different inputs; the live demonstration uses the interface profile above. No visits, negotiations or buyer decisions were observed in this exercise.

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12. Limitations

12.1 Data Limitations

- **Historical Listings**: No verification of current availability
- **Limited Geocoding**: Only 133/6,340 catalog listings (2.1%) have precise coordinates
- **Maintenance Data**: 4,453 of 6,340 catalog listings (70.24%) lack maintenance fee information; 2,009 belong to the historical base
- **Temporal Validation**: Cannot assess price trends or market changes
- **Selection Bias**: 91 listings excluded by IQR filtering

12.2 Model Limitations

- **High-Price Segment**: Poor performance on luxury properties (≥ S/ 2M)
- **District Coverage**: Limited to 20 districts in dataset
- **No Temporal Validation**: Cannot assess model degradation over time
- **Feature Limitations**: Does not include property condition, age, or amenities beyond basic specs

12.3 Product Limitations

- **Local Only**: Prototype runs on localhost; not deployed
- **SQLite Persistence**: Not suitable for production (single-user, file-based)
- **No Email Verification**: Registration requires email but no verification sent
- **No Real-Time Updates**: Catalog is static; no live data integration
- **No Contact Integration**: Cannot contact sellers directly through the platform

12.4 Legal and Ethical Limitations

- **Data License**: Urbania data is for academic demonstration only
- **No Commercial Use**: Cannot be used for commercial real estate transactions
- **No Financial Advice**: Affordability calculations are simulations, not pre-qualification
- **Privacy Considerations**: User data stored locally; no GDPR compliance assessment

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13. Conclusions

13.1 Achievements

InmoScore successfully demonstrates a data product that:

- Addresses a real problem (home buying complexity)
- Integrates ML (price prediction) with user preferences
- Provides transparent, explainable recommendations
- Works reliably across multiple use cases
- Has been rigorously evaluated (22% error reduction vs. baseline)

13.2 Key Learnings

1. **Personalization Matters**: Users value rankings tailored to their financial situation
2. **Transparency Builds Trust**: Explaining recommendations increases user confidence
3. **Simplicity Wins**: Complex ML needs simple, accessible UI
4. **Data Quality is Critical**: Historical data has inherent limitations
5. **Prototype-First Approach**: Iterative refinement from Week 7 to Week 15 improved product significantly

13.3 Future Work

Short-term (immediate improvements):

- Deploy to production with PostgreSQL/RDS
- Add email verification for registration
- Implement password recovery
- Add more districts to dataset
- Improve geocoding coverage

Medium-term (enhanced features):

- Real-time data integration with real estate APIs
- Temporal validation with time-series analysis
- Mobile app (React Native or PWA)
- Enhanced price prediction with image analysis
- Social features (share listings, ask friends)

Long-term (research directions):

- Multi-city expansion (beyond Lima)
- Investment property analysis (rental yield calculations)
- Neighborhood gentrification prediction
- Integration with mortgage pre-qualification
- Chatbot for natural language property search

13.4 Final Assessment

InmoScore achieves its core objective: helping home buyers in Lima make more informed decisions by providing personalized, transparent, and data-driven property recommendations. While the prototype has limitations (historical data, local deployment, SQLite persistence), it demonstrates a viable approach to combining machine learning with user-centric design to solve a real-world problem.

The product is a local academic prototype. A production phase requires verified inventory, a deployment and persistence plan, and further user, security and operational validation.

Demonstration video

Pending recording and publication. No published URL is available yet. The team must record the 5-7 minute demonstration described in `DEMO_VIDEO_INSTRUCTIONS.md` and add the verified URL to this report and the project README.

Appendix

A. Reproducibility Instructions

```
# Install dependencies
npm ci

# Run development server
npm run dev -- --hostname 127.0.0.1 --port 5173

# Run tests
npm test
npm run test:api

# Validate data
python3.12 -m venv .venv
.venv/bin/python -m pip install -r analytics/requirements.txt
.venv/bin/python analytics/validate_data.py
```

B. File Structure

```
final_week/
├── app/                    # Next.js application
├── components/             # React components
├── lib/                    # Core logic (catalog, recommendation)
├── db/                     # Database layer
├── analytics/              # Data preparation scripts
├── models/                 # Trained ML models
├── data/                   # Catalog and processed data
├── tests/                  # Test suites
├── public/                 # Static assets
├── FinalReport.md          # This document
├── README.md               # Setup and execution instructions
├── DEMO.md                 # Demonstration guide
├── PresentationFinal.pptx  # Final presentation
└── package.json            # Dependencies
```

C. References

- Dataset: `MathiuCz/lima-real-estate-price-predictor` (GitHub)
- Framework: Next.js 16.3.4, React 19
- ML Library: scikit-learn
- Authentication: Custom scrypt implementation
- Icons: Lucide React

End of Final Report